Experiment 1

Sensitivity to changes in stimulus intensity

Blueprint Analytics

11 Oct 2018

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This script is part 5 of our analysis of the stimulus-response characteristics of the SPARS. In this analysis we examined whether there is a linear relationship between change in stimulus intensity and change in rating magnitude between two successive stimuli, irrespective of the direction of the change?

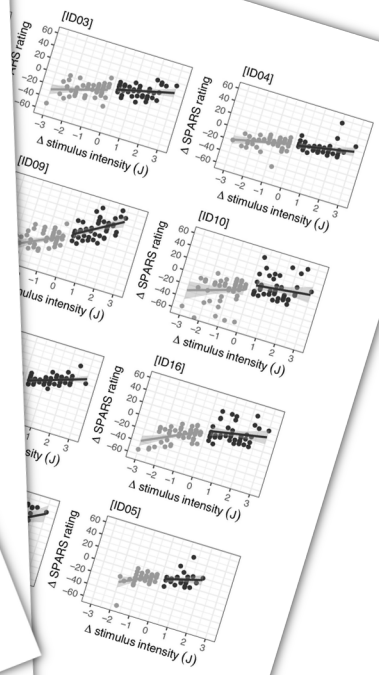
Descriptive plots of the data are provided in *"outputs/supplement_5.pdf"*, modelling of the stimulus-response relationship is described in *"outputs/supplement_6.pdf"*, the diagnostics on the final linear mixed model are described in *"outputs/supplement_7.pdf"*, the stability of the model is described in *"outputs/supplement_8.pdf"*, and the variance in ratings at each stimulus intensity is described in *"outputs/supplement_10.pdf"*.

Import and clean/transform data

```
##########################################################
#                                                        #
#                        Import                          #
#                                                        #
##########################################################
data <- read_rds('./data-cleaned/SPARS_A.rds')

##########################################################
#                                                        #
#                        Clean                           #
#                                                        #
##########################################################
data %<>%
  # Select required columns
  select(PID, block, block_order, trial_number, intensity, intensity_char, rating)
```



```
intensity change',
),
+323232')) +

o = 3, by = 1),

= 60, by = 20),
```



was affected
or less than
'up' or 'down'):
interaction term

$$\text{ction of change} + (1 \mid participant)$$